PLANWISE ENTERPRISE

Computational Development Platform

From a piece of land to a build-ready project.

Land-to-Home Development & Construction Platform

Unifying statutory land feasibility, computational house design, model-linked quantities, regional cost waterfalls, and licensed professional review into one traceable, deterministic workflow.



MARKET REALITY

Real estate development is fragmented.

A single residential parcel passes through disconnected silos, causing repeated rework, cost drift, and project delays.

THE TRADITIONAL SEQUENTIAL RELAY (HIGH FRICTION & DATA RE-CREATION)

Land → Survey → Feasibility → Architect → Design → BOQ → Costing → Engineer → Approvals → Contractor → Construction

⚠ Every handoff occurs between disjointed software tools, PDFs, and spreadsheets with zero shared single source of truth.

PROBLEM 01

Disconnected Data Silos

Data gets recreated repeatedly

- Surveys, CAD drawings, and spreadsheets live in isolated files across separate consulting firms.
- Every stakeholder manually redraws project geometry from scratch, introducing discrepancies.
- No single traceable project model connects land coordinates to construction handoff.

PROBLEM 02

Rework & Cost Drift

Design adjustments break downstream cost

- Statutory feasibility is largely manual and error-prone during early land evaluation.
- Downstream BOQ and cost estimates drift continuously from iterative design modifications.
- Cost blindspots and scope mismatches are discovered too late, triggering painful redesigns.

PROBLEM 03

Delayed Validation

Professional review happens too late

- Licensed structural and statutory compliance reviews only occur after design freeze.
- Non-compliant designs require costly re-drafting and delay municipal approval cycles.
- Customers and builders struggle to interpret opaque technical and engineering decisions.

CORE INNOVATION

One computational workflow. One source of truth.

The Canonical Building Model (CBM) links every stage from land parcel coordinates to licensed build release.

THE UNIFIED PLANWISE WORKFLOW

01 Land Intake
SECS topocentric frame & boundary survey

02 Statutory Feasibility
Mumbai DCPR 2034 rules, setbacks & envelope

03 House Generation
Multi-constraint CP-SAT parametric layouts

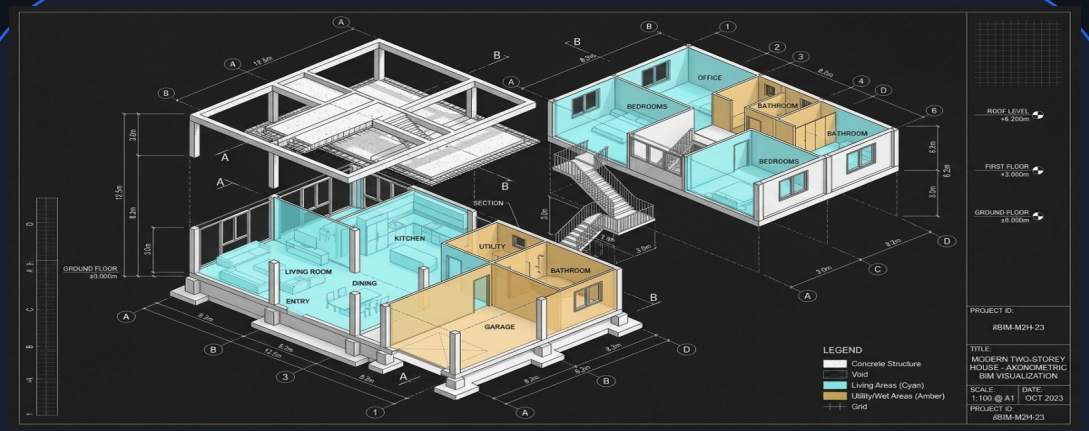
04 Canonical Building Model
Semantic 3D single source of truth (CBM)

05 Model-Linked QTO
Automated IS 1200 rules & geometric deductions

06 Cost Waterfall
Regional trade, material, labour & margin breakdown

07 Professional Review
G0–G6 multi-discipline engineer review ledger

08 Build Release
Cryptographic SHA-256 seal & IFC4 BIM package



THE CANONICAL BUILDING MODEL (CBM) DIFFERENTIATOR

The same underlying mathematical building model simultaneously drives:

- ✓ Floor plans & 2D/3D geometry without manual drafting
- ✓ Automated 12-point statutory & physical geometric validation
- ✓ Model-linked material takeoff (concrete, AAC blocks, plaster)
- ✓ Hierarchical 7-trade BOQ tied directly to 3D element IDs
- ✓ Live regional construction cost waterfalls
- ✓ Multi-discipline licensed engineer review and tamper-evident release

PLATFORM MODULES

What Planwise actually does

Six integrated computational modules replacing weeks of fragmented pre-construction consulting.

MODULE 01

Land Intelligence

Reconstructs surveyed parcel boundaries into an authoritative topocentric SECS coordinate frame with geodetic GPS anchoring.

MODULE 02

Feasibility Engine

Applies statutory municipal rulepacks (e.g. Mumbai DCPR 2034) to calculate road widening, setbacks, FSI, and buildable envelopes in seconds.

MODULE 03

Computational Design

Solves multi-constraint topological layouts using Google OR-Tools CP-SAT to produce Pareto-optimal, compliant residential floor plans.

MODULE 04

Quantity & Cost Engine

Extracts exact material volumes directly from 3D geometry using IS 1200 deduction rules, linked to regional trade rate snapshots.

MODULE 05

Professional Review

Enforces a structured G0–G6 verification ledger for certified land surveyors, architects, and structural engineers.

MODULE 06

Build Handoff

Compiles tamper-evident IFC4 BIM files, validation audit reports, itemized BOQs, and cryptographic SHA-256 release certificates.

CUSTOMER JOURNEY

Complex engineering underneath. Simple experience on top.

Landowners, builders, and developers navigate a clear, transparent decision journey without technical friction.

STEP 01
Tell us about your land

Input coordinates, CTS revenue number, or upload registered surveyor boundary sketch.

STEP 02
See what can be built

Instant visual statutory envelope, permissible FSI limits, and municipal setback overlay.

STEP 03
Choose your house

Explore multiple Pareto-optimal floor plans and 3D architectural massing alternatives.

STEP 04
See estimated cost

Live, transparent preliminary cost breakdown by trades, materials, labour, and prelims.

STEP 05
Request review

Engage certified engineers and surveyors for structured multi-discipline gate verification.

STEP 06
Move to build

Receive stamped engineering audit, IFC4 BIM model, and contractor-ready 7-trade BOQ.

THE USER EXPERIENCE PROMISE
Replaces 4–8 weeks of fragmented consulting with minutes of computational clarity, giving property owners total transparency before committing capital.

VERIFIED CAPABILITIES

This is already a working platform.

Planwise is not an unbuilt concept—the core end-to-end computational pipeline is fully implemented.

Land, Law & Geometry

- ✓ SECS geodetic-to-planar coordinate transformation
- ✓ Site evidence engine with surveyor licence tracking
- ✓ Mumbai DCPR 2034 statutory rule engine (road widening & setbacks)
- ✓ Customer design brief intake (hard & soft constraints)
- ✓ CP-SAT parametric layout generator (Google OR-Tools)
- ✓ Canonical Building Model (CBM) semantic 3D data structure
- ✓ 12 automated statutory & physical geometric validation checks

Quantities, Cost & Verification

- ✓ Model-linked QTO takeoff with IS 1200 deduction rules
- ✓ Hierarchical 7-trade BOQ compilation
- ✓ Regional cost waterfall engine (INDIA-MUMBAI-2026-Q4-V1)
- ✓ Build Review customer acknowledgement workflow
- ✓ Engineer Review Ledger with G0–G6 multi-discipline gates
- ✓ Cryptographic SHA-256 release fingerprinting
- ✓ Export to IFC4 BIM, JSON manifests, and CSV takeoffs

Operational Evidence & UI

- ✓ Full-stack application: FastAPI gateway + React/TypeScript UI
- ✓ Interactive CAD parcel workspace with snapping & ortho tools
- ✓ Mapbox GIS workspace with cadastral overlays
- ✓ 131 backend unit & regression tests passing
- ✓ 20 Vitest geometry engine tests passing
- ✓ PostgreSQL schema with immutable release ledgers
- ✓ Controlled single-project pilot executed end-to-end

*Note: While core feasibility, generation, QTO, costing, and review are fully functional, physical construction authorization remains strictly gated for licensed professionals.

PILOT VERIFICATION

From software to a controlled real-project workflow.

Executed end-to-end on an authoritative 1,100 sq ft residential parcel in Bandra West, Mumbai.

AUTHORITATIVE PLOT AREA
1,100.0 sq ft
102.19 m² preserved invariant

LOCATION & RECORD
Bandra West, Mumbai
CTS-1842-BANDRA (DCPR 2034)

SELECTED DESIGN OPTION
Option B (G+1)
98.16 m² Gross BUA (76.8 m² Carpet)

PRELIMINARY COST WATERFALL
₹13,18,391 INR
Direct preliminary construction cost

12-STAGE CONTROLLED PILOT EXECUTION TRACE

- Reconstructed planar SECS boundary: 27.5 ft front × 40.0 ft depth.
- Deducted 3.0m road widening strip along southern road frontage.
- Applied Mumbai DCPR 2034 setbacks: 49.06 m² buildable envelope.
- CP-SAT solver generated 3 Pareto-optimal layouts; Option B selected.
- Compiled semantic CBM 3D model with 12/12 validation checks PASSED.
- Derived IS 1200 QTO: Concrete 29.45 m³, AAC Blockwork 38.12 m³.
- Computed 7-trade BOQ and regional cost waterfall using Q4 2026 rates.
- Persisted immutable release package with SHA-256 fingerprint.

PROFESSIONAL GATE AUDIT LEDGER (G0–G6)

[G0] Site & Boundary Verification: VERIFIED — By Registered Surveyor SURV-MH-4421
[G1] Regulatory Feasibility Review: VERIFIED — Mumbai DCPR 2034 setback & FSI compliance
[G2] Structural Grid Review: VERIFIED — IS 456 RC column alignment & spans verified
[G3] MEP & Services Review: VERIFIED — Vertical wet-wall stacking verified
[G4] Cost & BOQ Review: VERIFIED — IS 1200 deductions & regional trade rates
[G5] Municipal Build Authorization: LOCKED — Requires formal municipal building submission
[G6] Contractor Construction Handoff: LOCKED — Requires executed commercial contract

Controlled single-project pilot successfully executed. Municipal submission (G5) and construction handoff (G6) remain authority-controlled.

SYSTEM INTEGRITY

Built for traceability, not just visualization.

AI assists with customer preferences; deterministic engineering algorithms and licensed professionals govern the truth.

THE DETERMINISTIC TRACEABILITY HIERARCHY

Survey Evidence → Statutory Rules → Canonical Model (CBM) → Validation Checks → Quantities → Cost → Engineer Ledger → Release

WHAT ASSISTIVE AI DOES

Acceleration, natural input, and presentation

- Interprets natural language customer design briefs & preferences
- Assists with spatial adjacency suggestions and preliminary zoning
- Accelerates 3D visualization rendering and aesthetic styling
- Generates contextual summaries of complex statutory clauses

WHAT DETERMINISTIC RULES & ENGINEERS GOVERN

Absolute physical, legal & financial invariants

- ✓ Statutory setback distances, road reservations, and FSI caps
- ✓ Watertight 3D geometry topology and NBC 2016 room dimensions
- ✓ Structural column alignment, spans, and vertical wet-wall stacking
- ✓ Model-linked material takeoff calculations (IS 1200 deductions)
- ✓ Regional trade rate snapshots and construction cost waterfalls
- ✓ Licensed engineer sign-off and construction authorization (G0–G6)

Principle: AI never writes authoritative statutory setbacks, structural dimensions, quantities, or construction sign-offs.

COMPETITIVE ARCHITECTURE

Why Planwise is different

A structural comparison between Planwise, generative AI tools, and traditional fragmented consulting.

Platform Dimension	Planwise Enterprise	Generative AI Tools	Traditional Workflow
Land Feasibility	✓ **Algorithmic statutory rulepack**	— Ignored / No concept of land	✓ Manual & slow (2–4 weeks)
Rule-Aware Design	✓ **Multi-constraint CP-SAT solver**	— Unconstrained prompt/pixels	✓ Manual architectural drafting
Canonical 3D Model	✓ **Single 3D source of truth (CBM)**	— None (2D flat raster images)	— Fragmented CAD/BIM files
Model-Linked QTO	✓ **Automated IS 1200 deductions**	— None	Manual spreadsheet takeoffs
Traceable Costing	✓ **Live regional rate waterfall**	— None	Manual estimating / guesswork
Professional Gates	✓ **Integrated G0–G6 audit ledger**	— None	Separate, delayed reviews
Immutable Release	✓ **Cryptographic SHA-256 seal**	— None	— Uncontrolled file versions
Customer Experience	✓ **Transparent interactive web app**	✓ Simple text prompt	— Opaque technical documents
Construction Path	✓ **Exportable IFC4 BIM & BOQ**	— Pure artistic concept	✓ Disjointed trade handoffs

Planwise is not an image generator or CAD viewer; it is a full-stack computational development engine.

COMMERCIAL STRATEGY

How Planwise can become a business

A pragmatic B2B / B2B2C business model combining transaction fees with recurring software subscriptions.

REVENUE STREAM 01

Per-Project Release Fee

TRANSACTION / USAGE

Tiered project release fee charged upon generating verified project packages, stamped IFC4 BIM models, and tender-ready BOQs.

- Paid by landowners, builders, or project sponsors
- Low barrier to entry for one-off residential projects
- Scales directly with total development volume

REVENUE STREAM 02

Professional SaaS

B2B RECURRING SEAT

Seat-based monthly/annual workflow subscription for architectural practices, structural engineers, and registered land surveyors.

- Dedicated review ledgers & rulepack customizations
- Increases professional review throughput by 5–10x
- Sticky professional toolchain integration

REVENUE STREAM 03

Builder Enterprise Tier

B2B ANNUAL LICENSE

Annual enterprise software license for residential builders and real-estate plotting developers managing ongoing project pipelines.

- Unlimited parcel feasibility screenings
- Standardized house typology catalogues
- Direct procurement & supplier integration

REVENUE STREAM 04

Computational API

INFRASTRUCTURE LICENSING

Headless API licensing for GIS portals, mortgage pre-qualification platforms, and property tech networks.

- Instant setback, FSI, and cost estimation APIs
- High-volume, automated developer queries
- High-margin, low-touch software distribution

Commercial model aligns incentives: Developers save pre-construction time; professionals expand billable review capacity.

CUSTOMER SEGMENTS

Who pays and why

A segmented breakdown of paying customers across the residential development ecosystem.

PRIMARY CUSTOMERS
Immediate Go-to-Market Focus

Residential Builders
Evaluate land parcels in hours instead of weeks; generate accurate initial cost envelopes before committing capital.

Property Developers
Compress project kickoff cycles, eliminate manual re-drafting, and standardize multi-plot developments.

Architectural Practices
Automate tedious statutory zoning and schematic drafting, focusing billable hours on high-value creative detailing.

Structural Engineering Firms
Receive pre-coordinated geometric models with verified site boundaries, enabling efficient paid gate reviews.

SECONDARY CUSTOMERS
Growth Tier & Construction Partners

General Contractors
Receive verified 7-trade BOQs tied directly to 3D model elements, drastically reducing bid disputes and variation claims.

Real-Estate Advisory Firms
Instant zoning, FSI, and feasibility assessments to accelerate high-value land brokerage transactions.

Housing Plotting Networks
Scale standardized residential typologies across large plotted land subdivisions.

FUTURE CUSTOMERS
Enterprise Scale & Institutions

Proptech Portals & Networks
Embed automated feasibility and development intelligence directly into property listing platforms.

Housing Authorities & Municipalities
Automated regulatory compliance pre-checking for municipal building permit submissions.

Industrialized & Modular Builders
Direct computational data feed into offsite pre-cast and modular manufacturing factories.

VALUE PROPOSITION

Why would a professional organization pay?

Measurable operational efficiency, risk mitigation, and transparency for every stakeholder.

FOR THE BUILDER & DEVELOPER
Capital Efficiency & Risk Reduction

- Accelerates land evaluation from weeks to hours before committing land acquisition capital
- Eliminates early-stage cost blindspots with model-linked regional rate snapshots
- Standardizes pre-construction development workflows across distributed project teams
- Drastically reduces downstream change orders and construction contractor claims

FOR THE ARCHITECTURAL PRACTICE
Throughput & Workflow Modernization

- Replaces manual setback drafting and statutory zoning calculations with automated rulepacks
- Generates mathematically sound starting baselines (CBM) ready for creative refinement
- Eliminates tedious re-drawing when clients modify basic spatial or budgetary preferences
- Expands firm design capacity without proportionally increasing drafting overhead

FOR THE STRUCTURAL & MEP ENGINEER
Traceability & Controlled Governance

- Receives pre-coordinated geometric models with verified geodetic site boundaries
- Enforces explicit professional review gates (G0–G6) backed by immutable audit trails
- Verifies RC column grids and vertical wet-wall stacking before detailed calculations
- Protects professional liability with clear, tamper-evident sign-off records

FOR THE LANDOWNER & CUSTOMER
Transparency & Informed Decisions

- Demystifies complex municipal bylaws and construction planning into clear choices
- Guaranteed consistency: The selected floor plan strictly drives material quantities and cost
- Avoids contractor budget shocks by establishing an audited preliminary cost baseline
- Provides a clear, verified roadmap to licensed construction execution

COMPETITIVE MOAT

The moat is the system, not the AI image.

Defensibility is rooted in the tightly coupled, version-controlled computational graph linking physical land to cost.

THE CONNECTED COMPUTATIONAL GRAPH

Site Intelligence + Statutory Rules + CP-SAT Solvers + Canonical 3D Model (CBM) + IS 1200 QTO + Cost Engine + Review Ledger

A change in any layer propagates mathematically across every downstream artifact in milliseconds.

DEFENSIBILITY PILLAR 01

Coordinated Change Propagation

Physical & mathematical binding

- A 0.5m modification to a site boundary instantly cascades through statutory setbacks, CP-SAT room envelopes, 3D geometry, concrete quantities, and cost waterfalls.
- Standalone CAD, BIM, or estimating software cannot propagate changes across disciplines automatically.
- Prevents costly human errors and synchronization gaps.

DEFENSIBILITY PILLAR 02

Deterministic Integrity

Auditable engineering proofs

- Unlike probabilistic AI image generators that hallucinate impossible layouts, Planwise relies on formal constraint satisfaction (Google OR-Tools).
- Every calculation produces an auditable geometric and arithmetic proof suitable for licensed engineers.
- Creates legal and regulatory defensibility that builds institutional trust.

DEFENSIBILITY PILLAR 03

Institutional Rulepack Moat

Accumulating regional intelligence

- Local municipal bylaws (e.g. Mumbai DCPR 2034) and regional trade rate snapshots build an expanding institutional data barrier.
- Expanding rulepacks create high switching costs once builders and developers standardize on Planwise.
- Network effects: More verified pilot projects enhance regional rate and rule accuracy.

Defensibility comes from connecting these layers into one versioned, auditable workflow.

PLATFORM EVOLUTION

Platform roadmap: From pilot to enterprise scale

Disciplined technological execution transitioning from core engine verification to multi-market expansion.

PHASE 1: NOW
Implemented & Verified

- ✓ SECS planar land intake & geodetic GPS anchoring
- ✓ Mumbai DCPR 2034 statutory feasibility rule engine
- ✓ CP-SAT parametric 2D/3D house layout generator
- ✓ Canonical Building Model (CBM) semantic 3D schema
- ✓ Model-linked QTO takeoff (IS 1200 measurement rules)
- ✓ Regional cost waterfall engine (INDIA-MUMBAI-2026-Q4-V1)
- ✓ G0–G6 Engineer Review Ledger & SHA-256 release package
- ✓ Controlled 1,100 sq ft single-project pilot completed

PHASE 2: NEXT
Professional Pilots & Ecosystem

- → Interactive client walkthroughs & 3D WebGL model inspection
- → Multi-firm professional pilot cohort (builders, architects, engineers)
- → Municipal pre-check format exports (AutoDCR / CAD compliance)
- → Subcontractor procurement packages & digital tender generation
- → Mobile surveyor intake app with timestamped photo anchoring
- → Expanded regional cost database for Tier-1 Indian metros

PHASE 3: LATER
Multi-Typology & Enterprise Scale

- → Multi-typology expansion: commercial, mixed-use, row housing
- → National statutory rulepack library (NBC 2016, Delhi, Bengaluru)
- → Automated structural FEA integration & reinforcement schedules
- → On-site construction IoT progress tracking & digital twin syncing
- → Enterprise ERP integration for large real-estate development corporations

PARTNERSHIP OPPORTUNITIES

We are looking for the right partners.

Collaborating with forward-thinking builders, professionals, and strategic investors to scale computational development.

OPPORTUNITY 01

Builder & Developer Pilots

PILOT COLLABORATIONS

Forward-thinking residential builders and plotting developers seeking to pilot Planwise on upcoming real estate parcels for automated feasibility and costing.

OPPORTUNITY 02

Professional Review Network

ARCHITECTS & ENGINEERS

Registered architectural practices, certified structural engineers, and licensed land surveyors interested in joining our accredited review network.

OPPORTUNITY 03

Real Pilot Parcels

PROJECT SITES

Landowners, builders, and plotting developers with residential plots (800–2,500 sq ft) ready for feasibility screening, layout generation, and costing.

OPPORTUNITY 04

Technology & Industry Partners

ECOSYSTEM INTEGRATION

Collaborations with GIS data providers, BIM software ecosystems, municipal compliance systems, and construction technology platforms.

OPPORTUNITY 05

Strategic Capital & Advisors

INVESTMENT & GUIDANCE

Value-add strategic investors and real-estate industry leaders aligned with modernizing fragmented property development into a scalable computational science.

Focus: Quality pilot partnerships with active parcels and accredited professionals.

CALL TO ACTION

Let's build the next generation of development infrastructure.

Transforming fragmented real estate development into an integrated, computational science.

PATHWAY 01

Pilot Planwise

FOR BUILDERS & LANDOWNERS

Bring an active residential parcel to experience the end-to-end feasibility, design, QTO, and costing workflow on real site data.

→ **Schedule a 30-Minute Pilot Demonstration**

PATHWAY 02

Join the Network

FOR ARCHITECTS & ENGINEERS

Become a founding professional partner in our accredited review network for statutory compliance and structural sign-offs.

→ **Apply for Accredited Partner Network**

PATHWAY 03

Invest & Scale

FOR STRATEGIC INVESTORS

Partner with Planwise Enterprise as a strategic investor to modernise the residential development and construction software ecosystem.

→ **Request Detailed Investment Brief**

PLANWISE ENTERPRISE

Computational Real Estate Development & Construction Platform

Location: Mumbai, Maharashtra, India • Audience Outreach: Developers, Builders, AEC Professionals & Strategic Investors

Website: planwise.enterprise • Inquiries: partnerships@planwise.enterprise • Private & Confidential